

Pranav Ganesh Chandratre

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Education

University of Birmingham, Birmingham

(Sep 2025- Present)

MSc in Data Science

Currently enrolled for Sep 2025 Batch. As part of this course, I will delve into data science as new technologies reshape our world. I'll learn about contemporary challenges and cutting-edge techniques becoming an expert. Take part in Industry related project and gain on-hand experience.

Indian Institute of Science Education and Research, Kolkata

BS-MS Dual Degree Programme

(2017 - 2022)

Completed the integrated BS-MS in Physical Sciences (with specialization in Data Analysis and Machine Learning) 8.52 GPA.

Skills:

Programming languages: Python (*Matplotlib, Seaborn, NumPy, SciPy, Pandas, Scikit-Learn, Pytorch and Tensorflow* libraries), SQL, C, C++, MATLAB

Machine Learning Algorithms: Artificial Neural Networks, Reinforcement Learning, Convolutional Neural Networks, Graph Neural Network, Generative Adversarial Network, Transformer Networks, Diffusion Networks, Natural Language Processing, Agentic AI

Software: MS Office, Origin, LaTeX, CERN ROOT, Geant4 software

Work Experience

➤ MS thesis

(May 2021 – June 2024)

Particle identification in HGCal of the CMS Experiment using machine learning algorithms. [\[Report\]](#)

Indian Institute of Science Education and Research, Kolkata & Saha Institute of Nuclear Physics, Kolkata

- Developed supervised machine learning models for particle identification based on shower shapes for the high granularity calorimeter (**HGCal**) in the Compact Muon Solenoid (**CMS**) experiment, using machine learning algorithms, Artificial Neural Network, and Graph Neural Network, using a standalone GEANT4 simulation of a slice of detector.
- Achieved an accuracy of 94.76% for ANN and 96.66% GNN models.

➤ Post-MS Internship

After completion of my thesis in July 2022:

- Created multiple implementation of GNNs including Residual Networks and Attention Networks.
- Currently working towards adding noise simulations and improving model robustness.
- Improving the overall performance by integrating Residual networks for both ANN and GNN models.

➤ Projects Undertaken

Data Science Project: Predictive Modeling for UK Restaurant Insolvency

(Jan – March 2026)

Built an early-warning ML system to predict the failure of food and beverage establishments in the UK.

- Integrated 5.6M+ records across Companies House, FHRs, ONS, and Google Places API using complex fuzzy matching.
- Conducted zero-shot NLP sentiment analysis on customer reviews utilizing the Facebook/bart-large-mnli model.
- Resolved a severe class imbalance (5% minority class) using SMOTE and trained GPU-accelerated XGBoost and Logistic Regression models.
- Extracted feature importance, proving that structural metrics (e.g., company size) are superior predictors of liquidation over consumer sentiment.

Building CI/CD Workflows for Machine Learning Operations

(Dec 2024 – Jan 2025)

- Built end-to-end machine learning pipelines that will run when new data is pushed or code is updated.
- This is done using GitHub Actions, Gradio, Hugging Face, GitHub secrets.

Build and Deploy your Machine Learning Application

(Oct - Nov 2024)

- Used the Gradio Python library to build a Chabot application, and use the Hugging Face ecosystem to load the model and deploy it.

Analysis using Muon Scattering Tomography

(June –Sep 2023)

- Worked on creating a Convolutional Neural Network (U Net Image Segmentation Model) for localization and identification of civil structures and materials using data from Geant4 simulations from scattering of atmospheric muons.

A Study of Background Inflation Dynamics under Quadratic Scalar Potential.

(Aug - Dec 2021)

- Studied various problems that arise due to big Bang description of the universe and their proposed solution, inflation.
- Studied Inflation dynamics for a scalar inflation field and model a square potential and investigate the evolution of the field.

Environmental Kuznets Curve.

(Aug - Dec 2020)

- Verified the Environmental Kuznets Curve by exploring the correlation between the concentration of Nitrogen and Sulphur oxides and the GDP of different countries.
- Modelled the data according to a polynomial of order 2 according to the hypothesis.

Quantum Field Theory in Time Dependent Background

(Oct – Dec 2020)

- In this project, studied and demonstrated the expression for the scalar field and its governing equation in the FRW universe.
- Proceeded to quantize the field by invoking proper commutation relations.
- Briefly described of the vacuum state and the particle state corresponding to different mode functions.

Study the effect of Magnetic Force on Coupled Pendulums

(Oct – Nov 2018)

- Studied the effect of magnets on a coupled oscillator with iron bobs, modelled the result and studied the evolution of motion.

➤ Positions Undertaken

Poster Presentation:

(18th March 2023)

Indian Institute of Science Education and Research, Kolkata

- Presented a poster "Particle Identification based on Shower shapes in CMS using Machine Learning Algorithms" at Department of Physical Sciences, IISER Kolkata foundation day.

Teaching Assistant Position:

(Dec 2021 - April 2022)

Indian Institute of Science Education and Research, Kolkata

- I was the Teaching assistant for the first-year laboratory course in physics.

Achievements and Certifications:

- PwC(Google_powered) Langchain AgenticAI LAngsmith workshop:PwC[[Certificate](#)] **(February 2026)**
- Frontier Tech Hackathon **(March 2026)**
- Machine Learning Specialization: Coursera.org[[Certificate](#)] **(January 2023)**

- Convolutional Neural Networks: Deeplearning.AI[[Certificate](#)] (January 2023)
- Generative Adversarial Networks: Deeplearning.AI[[Certificate](#)] (May 2023)
- Sequence Models: Deeplearning.AI[[Certificate](#)] (February 2024)
- Generative AI with Diffusion Models: Nvidia[[Certificate](#)] (October 2025)
- INSPIRE Scholarship for Higher Education:
Department of Science and Technology (DST), Government of India (2017-2022)
- National Science Olympiad (NSO) Certificate of Merit: Science Olympiad Foundation
Secured 666th rank in round II (State Level). (November 2012)